MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE

EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCFSQ100 - INTRODUCTION TO FOOD SCIENCE

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) andAppreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. C) Food toxicants
2. B) Slows down bacterial growth without significantly changing the taste or texture of food
3. B) To ensure that food is safe and has a longer shelf life
4. B) Vitamin D
5. A) Physical hazards
6. C) Carbohydrate
7. A) Microbial growth
8. B) Buying in bulk
9. C) 7
10. C) Rice

Part B

Short Essay Type Questions

Answer any **4** Questions

Each Question carries **5** marks

11. Describe about a) Thermophilicbacterias b) Psychrotropicbacterias c) Halotolerantbacterias d) Aciduric bacterias

a) Thermophilic Bacteria

- Thrive at high temperatures (45–70°C).
- Commonly found in hot springs, compost, and industrial heaters.
- In food, they spoil canned foods, dehydrated foods, milk powders.
- Examples: *Bacillus stearothermophilus*, *Thermus* spp.
- Their spores survive heat → important in thermal processing.

b) Psychrotropic Bacteria

- Grow at low temperatures (0–7°C) but optimum at 20–30°C.
- Responsible for spoilage of refrigerated foods.
- Produce lipases and proteases that survive pasteurization.
- Examples: *Pseudomonas*, *Listeria monocytogenes*.
- Cause spoilage in milk, meat, fish, vegetables.

c) Halotolerant Bacteria

- Can grow in high salt concentrations, but do not require salt.
- Found in salted foods, cured meats, pickles, fish sauces.
- Example: *Staphylococcus aureus* (food poisoning bacteria).
- Important in food preservation, fermented foods.
- Spoilage risk in salted products if hygiene is poor.

d) Aciduric Bacteria

- Survive and grow in low pH (acidic) foods.
- Common in fruits, juices, pickles, fermented foods.
- Examples: *Lactobacillus*, *Acetobacter*, *Leuconostoc*.
- Responsible for spoilage of acidic beverages.
- Important in food fermentation (e.g., yogurt, sauerkraut).

12. Explain how food regulates various bodily activities

- Provides energy for all body processes.
- Proteins help growth, repair, and build enzymes and hormones.
- Vitamins and minerals regulate metabolic reactions.
- Fats support hormone production and temperature regulation.
- Minerals maintain fluid balance, nerve impulses, and muscle function.
- Nutrients support strong immunity.
- Dietary fiber regulates digestion and bowel movement.
- B-vitamins support brain and nervous system function.

13. Describe the role of world health organization in promoting global food safety

- Develops international food standards (Codex Alimentarius).
- Conducts global surveillance of foodborne diseases.
- Provides scientific risk assessments on contaminants and additives.
- Helps countries strengthen food safety laws and systems.
- Issues guidelines on hygiene and safe food handling.
- Promotes awareness through programs
- Coordinates international response to food safety emergencies.
- Supports training, research, and capacity building.

14. Write a note on minerals

- Needed in small amounts but vital for health.
- **Types of Minerals:**
- Macrominerals: Calcium, phosphorus, magnesium, sodium, potassium, chloride.
- Trace minerals: Iron, zinc, iodine, copper, selenium, fluoride.
- **Functions:**
- Bone formation: Calcium, phosphorus.
- Muscle contraction and nerve transmission: Sodium, potassium.
- Oxygen transport: Iron.
- Thyroid hormone formation: Iodine.
- Antioxidant defense: Selenium.
- **Sources:**
- Milk, leafy vegetables, nuts, meat, cereals, and fruits.
- **Deficiency Effects:**
- Iron deficiency: Anemia
- Iodine deficiency: Goiter
- Calcium deficiency: Osteoporosis
- **Toxicity:**
- Excess intake may cause health issues, e.g., high sodium → hypertension.

15. Mention the importance of protein in food

- Proteins help in growth and repair of body tissues.
- They form enzymes that control digestion and metabolic activities.
- They help in the production of hormones such as insulin.
- Proteins build antibodies and support the immune system.
- They help maintain fluid and electrolyte balance in the body.
- Proteins act as an energy source when carbohydrates and fats are insufficient.
- They transport and store important substances (example: hemoglobin transports oxygen).
- Proteins provide structure to muscles, skin, hair, and nails.

16. Evaluate and conclude the most controllable hazard in food with reasons

Food hazards are biological, chemical, and physical.Among these, physical hazards are the most controllable.

Reasons:

- Physical hazards like glass, metal, stones, and plastic pieces are easy to detect.
- They can be prevented through simple measures like proper cleaning, inspection, and equipment maintenance.
- Good Manufacturing Practices (GMP) such as sieving, filtering, and visual checking help control these hazards.
- Physical hazards do not multiply or grow over time, unlike biological hazards.
- Their prevention does not depend on temperature, pH, or moisture conditions.

Physical hazards are the most controllable because they are visible, easily preventable, and can be eliminated through basic quality-control practices.

Part C

Essay Type Questions

Answer any **2** Questions

Each Question carries **10** marks

17**. Discuss on the requirements for a successful entrepreneur**

- A successful entrepreneur must have a clear vision and long-term goals.
- Creativity and innovation are essential to identify new opportunities.
- Risk-taking ability helps the entrepreneur face uncertainties confidently.
- Strong decision-making skills allow quick and effective problem solving.
- Good communication skills are needed to lead teams and convince stakeholders.
- Financial management skills help in budgeting, cost control, and investment planning.
- Leadership qualities motivate employees and build a productive work environment.
- Market awareness helps understand customer needs and competition.
- Adaptability allows the entrepreneur to adjust to changing market trends.
- Persistence and self-confidence help overcome failures and continue working toward success.

18**. Write a note on fats**

- Fats are essential macronutrients that provide concentrated energy to the body.
- They supply 9 kcal of energy per gram, which is higher than carbohydrates and proteins.
- Fats help in the absorption of fat-soluble vitamins A, D, E, and K.
- They provide essential fatty acids such as omega-3 and omega-6, which the body cannot synthesize.
- Fats maintain body temperature by acting as an insulator.
- They form cell membranes and support normal cell function.
- Fats enhance the flavour, texture, and palatability of food.

- They serve as a protective cushion for internal organs.
- Moderate intake of healthy fats supports brain and hormone function.
- Excessive intake of saturated and trans fats leads to obesity, heart disease, and metabolic disorders.

19. **Explain Quality Assurance in food industry**

- Quality Assurance (QA) is a systematic approach to ensure food products meet safety and quality standards.
- QA focuses on preventing defects rather than detecting them after production.
- It involves the implementation of Good Manufacturing Practices (GMP).
- Standard Operating Procedures (SOPs) are developed and followed to maintain consistency.
- QA ensures compliance with national and international food regulations.
- Hazard Analysis and Critical Control Point (HACCP) is used to identify and control hazards.
- Regular inspections, audits, and documentation help maintain quality.
- QA includes training workers to maintain hygiene and follow safety protocols.
- It ensures traceability by keeping proper records from raw materials to finished products.
- QA builds consumer trust, reduces product recalls, and improves overall productivity.

20. **Explain about the causes of Food insecurity**

- Poverty is the major cause of food insecurity because people cannot afford nutritious food.
- Climate change leads to droughts, floods, and reduced crop production.
- Rapid population growth increases demand for food resources.
- Poor agricultural productivity affects the availability of food.
- Political instability disrupts food supply chains and access.
- Unemployment reduces purchasing power, increasing food insecurity.
- Natural disasters destroy crops, livestock, and infrastructure.
- Lack of access to land and water resources affects farmers' ability to grow food.
- Inefficient storage and transportation systems lead to post-harvest losses.
- Rising food prices make essential staples difficult for low-income families to buy.